

Sergio Córdoba, MSc.

Data | AI | Analytics

From data to business assets

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About me...

<https://sergiocy.github.io/>

With mathematical background and 10 years of expertise around data, i have an strong analytical (and business) thinking and technical capabilities, that i apply to get production-ready AI or ML systems in high-scale environments and data-driven companies, where there are an economic impact.

I have designed and deployed ML pipelines and AI systems in production, including segmentations, optimization processes, recommendation systems and RAG-based internal knowledge systems, owning the full lifecycle, playing a cross-functional role and engaging with stakeholders.

Languages...

Spanish; native

English; conversational

French; basic

Academic background

Postdegree in "Advanced methods in applied statistic" (UNED, Spain, 2017)

MSc. in "Mathematical modeling and research, statistics and computing" (University of Zaragoza, Spain, 2015)

Postdegree in "Microsystems and smart instrumentation" (University of Zaragoza, Spain, 2013)

MSc. in Physics (spanish 5-years-degree. University of Zaragoza, Spain, 2013)

Workshops and courses

#tensorflow #datarobot
#IBMWatson #neuralnetworks
#hadoop #spark #agile #linux
#simulation
#computationalscience
#softwarearchitecture

Experience

*** Senior/staff data scientist at Vueling Airlines / KnowMadMood, data&AI dept, 2018-2025, Zaragoza (Spain)

I have developed data products for distinct business areas (sales, customer, digital)

- engaging with stakeholders to understand the use-case and business goal
- reviewing state-of-the-art around the use-case to define the **model approach**.
- defining tasks and timelines and deliveries (**jira, scrum, crisp-dm, kanban**)
- performing **business presentations**; data descriptives, model results, business-cases, scenarios, simulations. **Reporting and ad-hoc analysis** (**powerpoint, matplotlib, seaborn, plotly, ggplot, excel, jupyter**)
- **Project ownership, tracking and monitoring** the model performance and "doing bridge" with business-areas to analyze and monetize the output.
- **Hybrid and cross-functional expertise** along data and cloud. **Applied ML-engineering**. ntegration with IT, data-engineering and MLOps teams.

I design, build and industrialize end-to-end ML or AI pipelines (training, predict) and i have strong proficiency in all stages.

- **data extraction and preparation**, feature engineering, cleaning, encoding, preprocessing. **ETL**, Strong in **SQL**, core in **AWS-Redshift + AWS-S3. Big data**.
- **Exploratory Data Analysis (EDA)**; features descriptives, business insights. Univariate, multivariate, correlations, interactions. **Python, pandas, numpy, matplotlib**
- **Modeling and evaluation**; sampling, stratification, feature selection, performance metrics (**F1, MAE, ROC**, etc). Supervised, unsupervised. **Sklearn, pandas, numpy, scipy**.
- **Deployment** (and exploitation) as **batch-process (AWS-ECR, docker, airflow)** and **API-REST (AWS-APIGateway, postman, flask, JSON, AWS-Lambda)** via.. Also, **AzureDevOps**.
- **Monitoring and maintenance**; logs, alerts, mailing, business KPIs (conversion-rate, revenue, RASK, yield, churn-rate). **python, dash, grafana, datadog, kibana, tableau**

Along the time i have been part of a team's growth and have faced different challenges technological and functional,

- * 2025 – **genAI, RAG, LLMs and LLMOps** (**Azure OpenAI, Azure DevOps**)
important RAG to load a huge quantity (~7000 pages) of company documentation
NPS-tracking and (PoC) Marketing Automation
- * 2023-2024 - **personalization tech-lead** (supervision and mentoring of 1 ds-junior)
Set of projects: **new customers datamart + binary classifier + clustering + segmentation + life-time value** (**xgboost, lightgbm, kmeans, kprototypes, multiclass**)
IT WAS REFERENCE OF CUSTOMERS PERSONAL INFO AND HISTORY
planning and high-level reporting
(**PoCs**) **integration of web activity** behaviour as segmentations input (**GCP, GA4, UA, GoogleAnalytics, CloudStorage, CloudFunctions, BigQuery**)
- * 2021-2022 - propensity to purchase from web-activity. **PoCs, analysis**
product-**recommender**; **integration DS + IT-Mobile** team, **AWS-Sagemaker**
- * 2020 – (bag) **pricing optimization** (**regression, demand, xgboost, ONNX, datarobot, A-B testing, price-elasticity**)
IT WAS THE FIRST MODEL DEPLOYED IN PRO IN PRICING PLATFORM
AN INCREMENT OF 10% OF REVENUE WAS ESTIMATED
IT WORKED IN PRODUCTIVE WEB PROVIDING LUGGAGE PRICES
- * 2019 – **demand forecast pipelines. Heterogeneous data sources** and code. **R-language, python, hive, pyspark, sparkR, spark, sparkSQL, hdfs, parquet**
involved in project standarizacion (project benchmarks, MLflow deployment)
- * 2018 – **Starting team** (i was first person at data dept in Zaragoza Office, distributed team, remote work); **Heterogeneous sources, systems and processes**.
Working over cloud-server (backend); **ssh, mobaxTerm, winscp, scp, linux, bash, AWS-EMR, AWS-EC2**
Anomaly-detection project; **R-language**, deployed over **AWS-EMR**. Scheduled and executed with **bash script and cron**. Later, orchestration with **MLflow**

*** PhD courses at UNED, 2018-2021, Madrid (Spain)

"Intelligent Systems" program. Around algebraic treatments of word-embeddings.
(python, R, nltk, BERT, Word2Vec, GloVe, text-similarity, LLM, research, wrtting)

*** Data scientist at NTT Data (everis), 2016-2018, Zaragoza (Spain)

Java, R-language, ontologies, semantic searcher, NLP, web-scraping, graphs data bases, SolR, elasticSearch, graphDB, OWL, RDF-storage, sparQL, pentaho, kibana